

A 10b/12b 40 kS/s SAR ADC With Data-Driven Noise Reduction Achieving up to 10.1b ENOB at 2.2 fJ/Conversion-Step

Pieter Harpe, *Member, IEEE*, Eugenio Cantatore, *Member, IEEE*, and Arthur van Roermund, *Senior Member, IEEE*

Abstract—This paper presents a power-efficient 10/12 bit 40 kS/s SAR ADC for sensor applications. It supports resolutions of 10 and 12 bit and sample rates from DC up to 40 kS/s to accommodate a variety of sensor applications. A Data-Driven Noise-Reduction method is introduced to selectively enhance the comparator noise performance. In this way, a higher ADC resolution can be achieved with a small increase of the power consumption. A self-oscillating comparator is used to generate the bit-cycling clock internally. In this way, the ADC only requires an external clock at the sample-rate frequency. A segmented capacitive DAC with 250 aF unit elements is applied to save power and to reduce DNL errors at the same time. The implemented prototype in 65 nm CMOS occupies an area of 0.076 mm². For the two supported resolutions (10/12 bit), the ADC achieves an ENOB of 9.4 and 10.1 bit while consuming 72 and 97 nW from a 0.6 V supply at 40 kS/s. This leads to power efficiencies of 2.7 and 2.2 fJ/conversion-step for 10 bit and 12 bit resolution, respectively. Furthermore, the leakage power, which is below 0.4 nW, ensures that the efficiency can be maintained down to very low sample rates.

Index Terms—ADC, analog-to-digital conversion, CMOS, successive approximation, noise reduction, comparator, majority voting.

I. INTRODUCTION

LOW-POWER sensor applications for e.g., environmental monitoring, bio-potential recording and wireless autonomous sensor networks require highly power-efficient ADCs. At present, most of these low-power ADCs are limited to effective resolutions below 10 bit. For instance, [1] and [2] achieve ENOBs of 9.1 and 9.3 bit with power-efficiencies of 3.2 and 2.8 fJ/conversion-step, respectively. While an ENOB below 10 bit can be sufficient for basic sensor applications, more sophisticated sensor systems typically employ ADCs with a resolution beyond 10 bit. For instance: biomedical applications (such as EEG and ECG monitoring [3]) and environmental applications (such as temperature, pressure and humidity monitoring [4]). However, current ADC designs with an ENOB of at least 10 bit are limited to power efficiencies beyond 10 fJ/conversion-step [5]–[9].

The aim of this work [10] is to increase the accuracy of highly-efficient SAR ADCs beyond 10 bit, while further improving the power-efficiency beyond present state-of-the-art (2.8 fJ/conversion-step, [2]). To do so, the main challenge is to deal in an efficient way with the thermal noise limitations since in high-resolution ADCs the power consumption is more driven by thermal noise constraints. In such a case, every 6 dB increase in resolution comes at the cost of a 4× increase in power consumption [11]. To overcome this fundamental scaling rule, this work introduces a Data-Driven Noise-Reduction method (DDNR) to efficiently suppress comparator noise by means of selective noise enhancement. Furthermore, a segmented charge-redistribution DAC with 250 aF unit elements is used for better power-efficiency and better DNL. Also, a self-oscillating comparator is implemented to generate the bit-cycling clock locally, such that only a clock at the sample-rate is required from outside.

The paper is organized as follows. Section II introduces the theory of the DDNR method, and Section III describes the design and implementation of the ADC. Measurement results are shown in Section IV and conclusions are given in Section V.

II. DATA-DRIVEN NOISE-REDUCTION METHOD

In high-resolution ADCs, which are limited by thermal noise, the power consumption of the analog components will increase by a factor of 4 for each additional 6 dB of SNDR (equivalent to 1 bit gain in ENOB). For that reason, it is difficult to maintain the excellent power-efficiency of SAR ADCs when extending the resolution beyond 10 bit. Considering the analog blocks inside the ADC, the comparator usually dominates the power consumption [12], [2]. In view of this, a method is presented here that scales down the comparator noise in a much more power-efficient way as compared to the conventional approach. Before this method is explained, a digital majority-voting technique is discussed as an alternative to analog scaling.

A. Majority Voting

Consider a comparator with a certain input-referred noise, denoted by its standard deviation σ_{noise} . Due to this noise, the comparator has a certain output probability function, as shown in Fig. 1, where P_0 denotes the probability of output ‘0’ as function of the input voltage and P_1 denotes the probability of output ‘1’ as function of the input voltage. Note that a differential implementation is assumed here, with a comparator reference voltage at $V_{\text{in}} = 0$ V. P_0 and P_1 are determined by the

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The authors are with Eindhoven University of Technology, Mixed-signal Microelectronics Group, 5612AZ Eindhoven, The Netherlands (e-mail: p.j.a.harpe@tue.nl).

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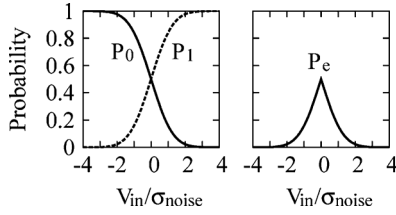


Fig. 1. Output probabilities (P_0 and P_1) and error probability (P_e) of the comparator.

Cumulative Distribution Function (CDF). Assuming a Gaussian distribution of the input-referred noise, the CDF is given by:

$$\text{CDF}(V_{\text{in}}) = \frac{1}{2} \left[1 + \text{erf} \left(\frac{V_{\text{in}}}{\sqrt{2}\sigma_{\text{noise}}} \right) \right] \quad (1)$$

and P_0 and P_1 are given by:

$$P_0(-V_{\text{in}}) = P_1(V_{\text{in}}) = \text{CDF}(V_{\text{in}}) \quad (2)$$

From P_0 and P_1 , the error probability P_e (i.e., the probability that the comparator takes the wrong decision) can be deducted as:

$$P_e(V_{\text{in}}) = \text{CDF}(-|V_{\text{in}}|) \quad (3)$$

For illustration, P_e is also shown in Fig. 1. As can be seen, P_e approximates zero for $|V_{\text{in}}| \gg \sigma_{\text{noise}}$, which logically means that the comparator makes no mistakes when the input signal is sufficiently large as compared to the noise level.

When the same comparator is used several times to repeat the same decision, a majority-voting logic can be applied on the obtained output samples to decide the final output. Using more samples increases the probability that the majority decision is correct. As an example, Fig. 2 shows the error probability of an initial comparator, as well as the performance after $5 \times$ voting ($P_{e,5v}$). For reference, the error probability using $4 \times$ analog scaling is also shown ($P_{e,4x}$). It can be observed that in this example the analog scaling and majority voting are similar in error performance. For the voting scheme, it is not always necessary to determine all 5 samples. E.g., if the first 3 votes are all equal to '0', the last 2 samples can be omitted as the majority is already clear, thereby saving power. In addition to the theoretical model, the statistical results from a transistor-level transient-noise simulation (including $1/f$ noise) of the actual comparator used in this work are also shown in Fig. 2. The comparator has an input-referred noise of $\sigma_{\text{noise}} = 305 \mu\text{V}$, which is reduced to $163 \mu\text{V}$ after $5 \times$ voting in agreement with the theoretical model. In this implementation, the comparator is a fully dynamic circuit. This ensures uncorrelated votes as potential memory effects and $1/f$ -noise are suppressed by the dynamic operation.

Analogous to the example for 5 samples, one can determine the error probability and the corresponding effect on σ_{noise} for any given number of samples. This is illustrated in Fig. 3, which confirms that voting on 5 samples will reduce the effective noise to about 50% of the original value, while voting on 25 samples will decrease it to about 25%, similar to $16 \times$ analog scaling. As

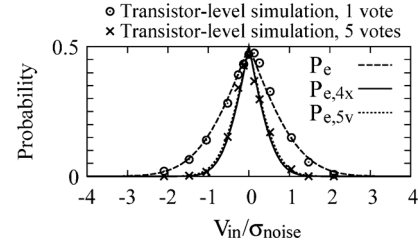


Fig. 2. Error probability of an initial comparator (P_e), after $4 \times$ analog scaling ($P_{e,4x}$) or after $5 \times$ majority voting ($P_{e,5v}$). Also shown are transistor-level transient-noise simulations of a comparator without voting and after $5 \times$ voting.

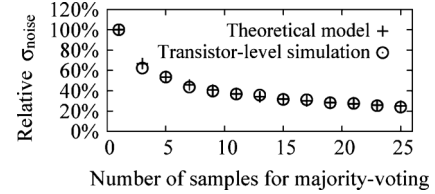


Fig. 3. Relative equivalent comparator noise as a function of the number of samples in the voting scheme, using the theoretical model and transistor-level transient-noise simulations.

can be seen, the theoretical model closely matches the transistor-level simulations.

In conclusion, the power consumption of the majority-voting scheme with 5 samples is similar to the power consumption of the equivalent analog-scaling method ($4 \times$), while the error probability is similar in both cases too. Also note that in an actual SAR ADC, the majority voting does not only reduce the impact of the noise from the comparator, but also reduces the impact of the noise from the feedback DAC. However, this work focuses on the comparator noise, which is more critical for the overall performance.

B. Data-Driven Operation

So far, the majority-voting procedure does not yet have a major impact on the achievable ADC performance. However, it offers flexibility as the amount of noise reduction can be dynamically adjusted simply by changing the number of samples used to determine the majority decision. This can be used advantageously as the noise requirement for the comparator is not identical for each bit-decision in the SAR algorithm: only in one comparison cycle is the amplitude of the input signal V_{in} to the comparator less than 0.5 LSB. Similarly, there is only one cycle in which $|V_{\text{in}}|$ is between 0.5 LSB and 1 LSB, one case where it is between 1 LSB and 2 LSB, and so on. Thanks to this known variation of the amplitude of V_{in} , the noise requirement for the comparator changes correspondingly: the requirement is most critical in one case only, while it becomes more and more relaxed for the other cases. This knowledge suggests that one could save power by adjusting the comparator noise performance depending on the amplitude of V_{in} . Thus, a technique is needed to provide flexible noise-performance, and another technique is needed to estimate the amplitude of V_{in} .

Flexible noise-performance can be implemented conveniently based on majority voting: the voting scheme can be instantaneously enabled in the noise critical case(s) to reduce

the noise, whereas the voting scheme can be disabled during the remaining cases to save power. To estimate the potential power savings of such a scheme, a 12 bit SAR ADC is considered as an example. To reduce the noise level of the comparator in such an ADC by $2\times$, permanent analog scaling would need $4\times$ scaling of the comparator, leading to a power-consumption increase of 300%. With a flexible majority-voting system, $5\times$ voting will be sufficient to get a similar noise reduction. If the voting is enabled only in one cycle, the total number of comparisons is $5 + 11 = 16$ instead of 12. Thus, the power increase is only 33% as opposed to the 300% of the conventional analog-scaling method.

The second requisite to implement the proposed method is a technique to estimate the amplitude of V_{in} , such that the most noise-critical cycle can be detected, which is not known beforehand. To automatically detect this, the comparator decision-time can be observed, as this decision time is inversely related to the amplitude of the input signal [13]. If the decision-time is longer than a certain threshold-time τ_{MV} , this implies that the input signal is smaller than a certain threshold voltage V_{MV} . In this way, the method can detect whether a decision is noise-critical or not. The threshold for being noise-critical can be set by tuning the delay τ_{MV} to achieve a desired V_{MV} . The actual hardware implementation of this circuitry will be discussed later in Section III.

The proposed method is denoted by the term Data-Driven Noise Reduction (DDNR), as it selectively applies noise reduction by means of majority voting, dependent on the input data (i.e., the amplitude information of V_{in}). The timing diagram of a 12 bit SAR conversion with DDNR is illustrated in Fig. 4: the comparator *Clk* initiates each comparison. Together with the result of the comparison, the comparator also generates a *Ready* indication as soon as the decision is made [2]. As shown by situation 1 in Fig. 4, if the *Ready* indication occurs fast enough, the comparator output is immediately accepted without majority voting as the input signal is apparently sufficiently large. In case the circuitry detects that the comparator decision is *too slow* (situation 2 in the figure), the decision may be corrupted by noise. Thus, the decision is postponed, 4 additional comparisons are performed, and majority voting is applied to the 5 samples to obtain the majority decision while suppressing the noise. The clock for the SAR logic normally follows the comparator *Ready* signal to control the bit-cycling operations. However, in case of voting, the SAR clock is suspended until the majority decision is reached and the next bit cycle can be activated.

C. DDNR Behavioral-Model Simulations

In this section, several simulation results on a behavioral model of a 12 bit SAR ADC with DDNR are discussed. The behavioral model in MATLAB includes an ideal 12 bit quantizer, complemented by a majority-voting scheme and an ideal amplitude-detector for selective noise reduction. Note that in this simplified behavioral model, the amplitude of the comparator input signal V_{in} is determined directly and compared against a threshold V_{MV} to decide whether or not the majority-voting scheme will be applied. As mentioned previously in Section II.B, the actual circuit implementation will do this detection indirectly through observing the comparator delay.

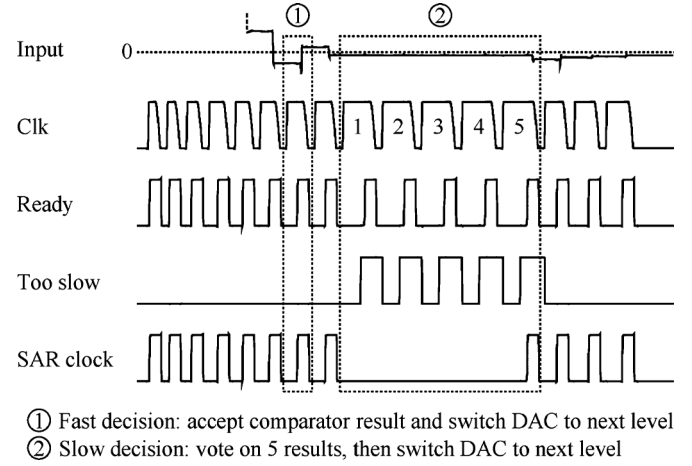


Fig. 4. Timing diagram of a 12 bit SAR conversion with Data-Driven Noise Reduction (DDNR).

Three noise sources are included during the simulations: quantization noise, sampling noise and comparator noise. The sampling noise is set to 0.289LSB , thus being equal to the quantization noise. The comparator noise (σ_{noise}) was set to a fraction α of the LSB:

$$\sigma_{\text{noise}} = \alpha \cdot \text{LSB} \quad (4)$$

where α was set to either $1/4$, $1/2$, or 1 during the simulations. Either 5 or 25 samples were used for majority-voting. For each case, a transient simulation was performed with a full-scale sine-input and with $1 \cdot 10^6$ data-points. From each simulation, the overall Input-Referred Noise (IRN) of the ADC was determined. Moreover, the average number of extra comparisons per 12 bit conversion was logged as it gives an indication of how much additional power consumption will be needed. The results are summarized in Fig. 5 as a function of the threshold voltage V_{MV} , here quantified in LSBs. In the IRN-plot, the horizontal lines without voting indicate the performance of a conventional ADC without DDNR. When voting is applied, the IRN will become dependent on the selected threshold level, as the threshold decides when to apply voting or not. For low thresholds (i.e., almost no voting will be applied), the DDNR ADC logically performs identical to a conventional ADC with the same comparator noise. When the threshold is being increased, the IRN will gradually go down until it reaches the limit as expected based on the number of samples in the voting scheme: with 5 or 25 samples, the noise reduces equivalent to $4\times$ or $16\times$ analog scaling, respectively. This is confirmed from the figure as the DDNR ADC with e.g., $\alpha = 1$ approximates the performance of a conventional ADC with $\alpha = 0.5$ when 5 votes are used and when the threshold setting is sufficiently high. Thus, it indeed performs similar to an analog design scaled by $4\times$. When 25 votes are used, the curve approximates a conventional ADC with $\alpha = 0.25$, equivalent to $16\times$ analog scaling. The circles in the figure indicate the points from whereon maximum noise reduction is obtained.

The second plot in Fig. 5 illustrates the average number of additional comparisons per conversion (on top of the 12 nominal ones), which logically increases as the threshold is moved

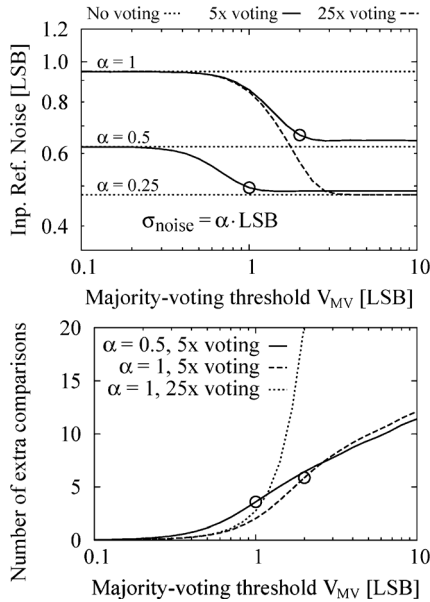


Fig. 5. Simulated ADC input-referred noise and number of extra comparisons versus majority-voting threshold for various amounts of comparator noise ($\sigma_{noise} = \alpha \cdot \text{LSB}$).

up. The circles in this plot (corresponding to the circles in the IRN plot) highlight that for 5 $\times$ voting and for $\alpha = 0.5$ or 1, about 4 or 6 additional comparisons are sufficient to get the maximum possible IRN-improvement. The number of additional comparisons is increasing with higher comparator noise (i.e., higher α), which can be understood from the fact that more decisions will become noise-critical when there is more comparator noise. Thus, the voting threshold needs to be moved up to allow proper noise suppression in case of increased comparator noise. Nonetheless, the 4 or 6 additional comparisons lead to an increase of about 30% or 50% of the overall power consumption only, which is substantially below the +300% required by conventional analog scaling. For the situation with voting on 25 samples, the power consumption (at maximum attainable IRN) increases by a factor of 5, which is at least 3 $\times$ more efficient than the equivalent 16 $\times$ analog scaling.

In literature, a few alternative solutions to efficiently enhance the noise performance of an ADC are known. For instance, [14] and [2] use redundancy in the search algorithm to save power for most bit-cycles at the cost of an additional cycle. Due to the redundancy, its theoretical power efficiency is lower as compared to this work. Besides, two separate comparators are needed (or a programmable one), which need to be matched or calibrated for offset. Digital correction logic is also required to map the internal bits with redundancy to a final binary output. On the other hand, the redundancy approach is advantageous for speed as it needs only a single additional cycle, while it can also compensate for DAC settling errors if sufficient redundancy is included. Another alternative, presented in [15], combines redundancy with averaging. While it can also correct for DAC settling errors, it requires more additional samples (8) as compared to this work due to the redundancy, which makes it a less power-efficient solution. Besides, additional DAC control logic and digital correction are required. As opposed to the alternatives men-

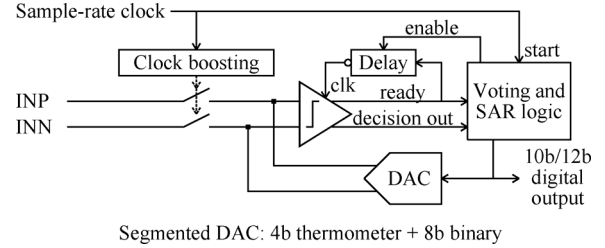


Fig. 6. SAR ADC architecture with segmented DAC and self-oscillating comparator.

tioned above, this work can detect the noise-critical bit cycle, therefore does not need redundancy, and is thus more power efficient.

III. SAR ADC IMPLEMENTATION

This section describes the implementation of a prototype SAR ADC with DDNR. Fig. 6 shows an overview of the implemented system. It operates at a single supply of 0.6 V, which is also used as the reference voltage for the DAC. To enable a large range of sensor applications, the ADC supports two resolutions (10 or 12 bit), and operates with sample rates from DC up to 40 kS/s. In 10 bit mode, the last two conversion cycles are simply omitted and the majority-voting logic is switched off to save power. Since the implemented circuits do not require static bias-currents, the power consumption will scale proportional to the sample rate. The bit-cycling clock is generated internally by forming an oscillation loop around the comparator. The DAC is implemented with a segmentation of 4 bit thermometer and 8 bit binary encoding. The logic implements the SAR and the majority-voting algorithms. The different components will be discussed in the following sections.

A. Segmented Charge-Redistribution DAC

The DAC is implemented as a 12 bit charge-redistribution DAC. A single-ended view is shown in Fig. 7; in reality the same structure is implemented twice to compose a pseudo-differential DAC. The 8 LSBs are binary-encoded ($B < 7:0 >$) whereas the 4 MSBs are thermometer-encoded ($T < 14:0 >$), as will be explained later. The differential rail-to-rail input signal is sampled on the capacitors of the charge-redistribution DAC through clock-boosted sampling switches (Fig. 6). The total capacitance of the DAC is 1 pF. As the input signal is directly sampled on this 1 pF capacitance, the SNR due to sampling noise is around 73 dB for a differential rail-to-rail input signal at 0.6 V supply. The unit capacitor for the LSB has to be as small as 250 aF to achieve an overall capacitance of 1 pF for 12 bit resolution. To implement such a small value, fringing capacitors are used as sketched in Fig. 7. To increase the capacitor density, metal layers 6 and 7 are used in parallel. As the capacitors are custom made, there is no mismatch information available from the foundry. However, based on the experimental data and modeling techniques in [16], it is predicted that the unit elements of 250 aF can have a matching with a standard deviation of 1%. This reasonably good matching for such a small capacitor is feasible as the equivalent capacitor area of $0.8 \mu\text{m}^2$ is relatively large.

From the theory in [17] and from behavioral-model simulations

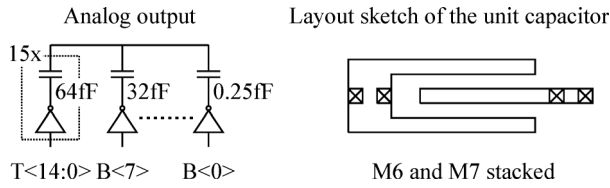


Fig. 7. Segmented charge-redistribution DAC.

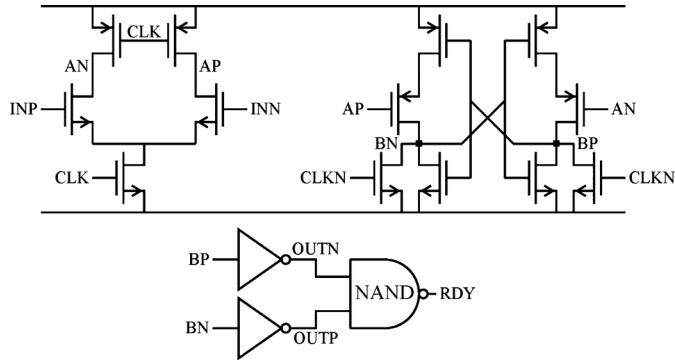


Fig. 8. Comparator implementation with "Comparison Ready"-output (RDY).

it is verified that 1% matching is sufficient to keep the INL and DNL within 1 LSB of a 12 bit ADC.

This work uses a segmented DAC in which the 4 MSBs are thermometer encoded while the remaining 8 LSBs are binary encoded [17]. The thermometer encoding reduces the probability of large DNL errors as typically present in binary-scaled architectures. A second advantage of thermometer encoding over a binary-scaled array is that it reduces the switching activity and thus the DAC switching energy [17], [10]. More complicated DAC structures, such as a split-capacitor array, were not necessary because the power of the DAC is not dominant in this implementation. To limit the complexity of the binary-to-thermometer-encoding logic and to limit the wiring overhead, only the 4 MSBs are thermometer encoded. From simulations it is verified that the encoder logic requires less than 2% of the overall ADC power. Besides, the binary encoding of the 8 LSBs has little impact on power or accuracy.

B. Self-Oscillating Comparator

Fig. 8 shows the implemented comparator, using a dynamic pre-amplifier followed by a dynamic latch [12]. The input-referred noise of the comparator is designed to be around 1 LSB in 12 bit mode, and thus equals 0.25 LSB in 10 bit mode. For 12 bit mode, majority voting on 5 samples will be employed to effectively reduce the comparator noise to 0.5 LSB. In 10 bit mode, the noise is already acceptable without majority voting.

A few logic gates are added to the comparator to generate an indicator *RDY* when the comparator reaches its decision. This ready indication is used both to control the majority-voting scheme and to generate an oversampled clock internally. A SAR ADC needs an oversampled clock to perform the *N* comparator operations, as well as to control the DAC and logic; furthermore this particular design will have a variable number of cycles per conversion because of the DDNR method. For simplicity of system integration, it is preferred to use a fixed sample-rate

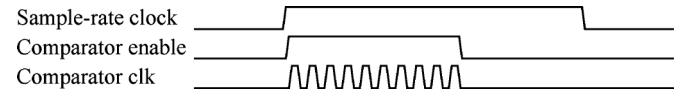
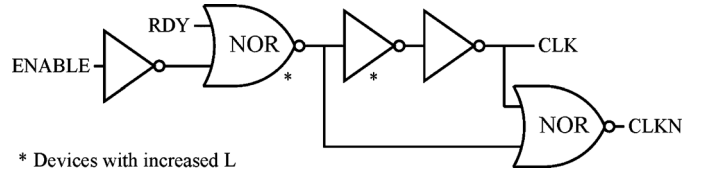


Fig. 9. Bit-cycling clock generation.



* Devices with increased L

Fig. 10. Self-oscillation circuitry around the comparator.

clock from the external world only. The required variable over-sampled clock is then generated inside the chip by means of an oscillation loop around the comparator [18]. This principle is summarized in Fig. 9: a rising edge of the external sampling clock initiates the conversion process. The comparator is now enabled, which also enables the self-oscillation of the comparator. The self-oscillation produces a clock signal that can be used to generate the timing for the SAR logic. In this way, standard synchronous CMOS logic can be used internally, even though the ADC as a whole is asynchronous with respect to the external clock. As soon as the last conversion cycle has been carried out, the comparator-enable signal is switched off such that the oscillation and clock generation are suspended until the next conversion will be initiated.

Fig. 10 shows the detailed implementation of the logic that generates the self-oscillation, and its behavior is given in Fig. 11. As long as the *ENABLE*-input is high, an inverted and delayed version of the comparison *RDY* signal is fed back to the comparator *CLK* input. In this way, the comparator starts self-oscillating and performs consecutive comparisons while generating the system clock at the same time. An additional NOR-gate is used to generate a non-overlapping *CLKN* which controls the latch-stage of the comparator. Further, several devices in the logic are intentionally made slower by increasing the length of the gates. This slows down the oscillation, such that there is sufficient time to reset the comparator, to allow for complete DAC settling, and to read out the data by the SAR logic. As the DAC settling is relatively fast thanks to the small capacitors (64 fF at most), and as the operation frequency of this ADC is relatively low (40 kS/s), it is straightforward to guarantee a sufficiently slow oscillation across all PVT corners.

C. Data-Driven Noise-Reduction Circuit

Fig. 12 shows the circuit implementation of the DDNR method. As mentioned before, a self-generated comparator clock *Clk* is created by a local feedback loop. At the same time, this *Clk* triggers a reference delay based on a current-starved inverter. In case the comparator is faster than the reference delay, the output decision and ready indication from the comparator are directly forwarded to the SAR logic. If the comparator is too slow, then no information is given to the SAR logic, but the majority voting logic is activated instead. In this prototype, a voting algorithm for 5 samples has been implemented: a first counter counts the number of comparator cycles, while a

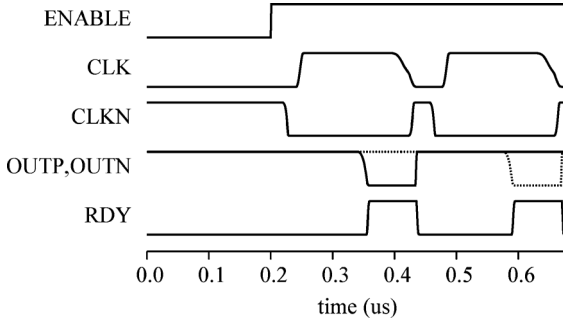


Fig. 11. Simulation of the self-oscillating comparator at 0.6 V supply.

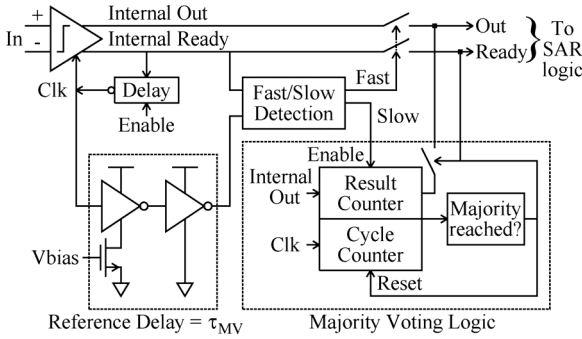


Fig. 12. Circuit implementation of the Data-Driven Noise-Reduction method (DDNR).

second counter counts the test results. As soon as the majority is clear, which takes between 3 and 5 votes, the majority value is produced as final output to the SAR logic and the voting logic is reset. The voting logic also creates a *Ready* pulse to activate the SAR logic. Note that the final output data towards the logic (*Out* and *Ready*) is identical regardless whether it comes directly from the comparator or from the majority voting logic. Thus, the SAR logic itself does not need any modification to support DDNR. The fast/slow detector is implemented by a gated SR-latch and the voting logic is implemented with standard digital logic.

As discussed in Section II.B, the delay of the reference circuit (τ_{MV}) determines a threshold voltage V_{MV} . If the input signal to the comparator is smaller than V_{MV} , the majority-voting scheme will be activated. In this prototype, the reference delay is externally controlled by hand through bias voltage V_{bias} (Fig. 12). For autonomous operation and reliability under PVT variations, a feedback loop could be implemented that controls V_{bias} based on the number of times the majority-voting is activated per conversion. Ideally this should happen e.g., once or twice per conversion, dependent on the desired performance improvement, the number of applied votes, and the comparator noise level (Fig. 5). By observing the number of activations in the digital domain, V_{bias} can be tuned to the optimum value.

IV. MEASUREMENTS

The ADC with a majority-voting scheme using 5 votes is implemented in 65 nm CMOS. It occupies an area of 0.076 mm² (Fig. 13), which includes the decoupling capacitors for the supply and reference voltage. It operates at 0.6 V supply and up to 40 kS/s. Figs. 14 and 15 show the measured INL and DNL,

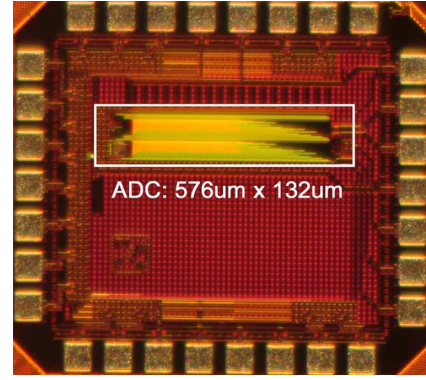


Fig. 13. Die photo of the ADC in 65 nm CMOS.

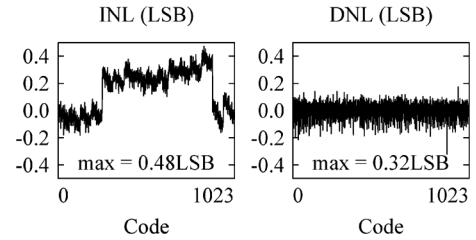


Fig. 14. Measured INL and DNL for 10b mode.

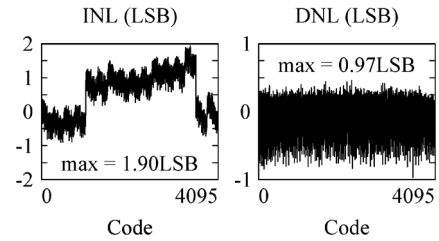


Fig. 15. Measured INL and DNL for 12b mode.

which remain below 0.48/0.32 LSB in 10 bit mode and below 1.90/0.97 LSB in 12 bit mode. While the 12 bit mode is not as good as expected, it is still a substantial improvement over our previous implementations [2], [16]. The two large errors in the INL pattern are most likely caused by systematic layout mismatch, as four measured test chips show an identical shape. Possibly, this systematic error is caused by the asymmetric capacitor wiring, the non-uniform metal density around the capacitors, or an insufficient amount of dummy elements. By determining the average INL of the 4 measured test chips, it is estimated that the INL is equally composed of systematic and random contributions. This implies that the INL could be within ± 1 LSB after correcting the systematic errors.

A near-Nyquist spectrum for 12 bit mode is shown in Fig. 16, and the measured ENOB versus input frequency is given in Fig. 17. The ENOB is 9.4 and 10.1 bit for 10 and 12 bit modes, respectively. The ENOB for 12 bit mode is mostly limited by the systematic INL error. Fig. 18 confirms that the measured power consumption scales with resolution and sample rate down to a leakage level of less than 0.4 nW. Thanks to this low leakage, the power consumption is only 1 nW when sampling in 12 bit mode with a rate of 250 S/s, which can be sufficient for quasi-static environmental monitoring or simple bio-potential recordings. According to post-layout simulations, 60% of the power is spent in

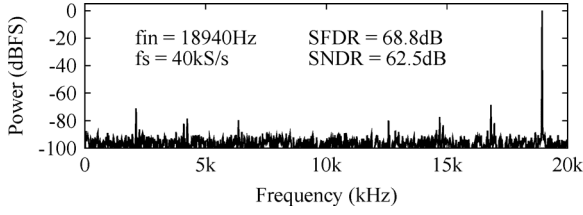


Fig. 16. Measured output spectrum for a near-Nyquist input tone in 12b mode.

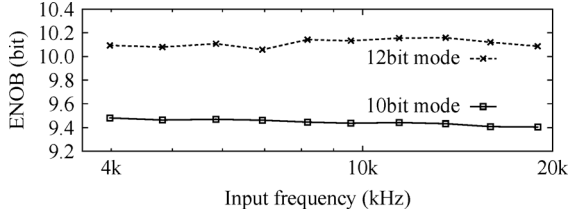


Fig. 17. Measured ENOB versus input frequency.

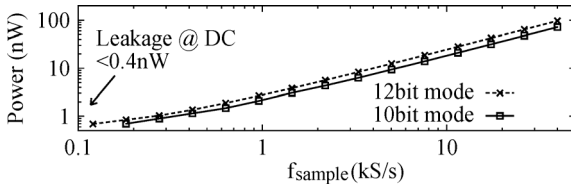


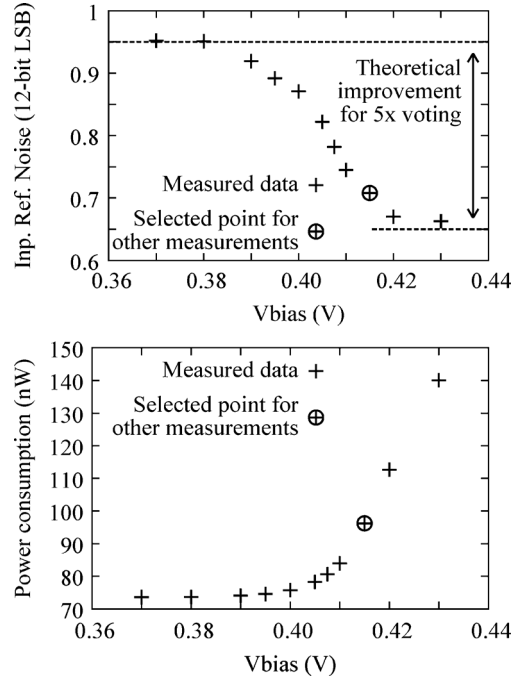
Fig. 18. Measured power consumption versus sample rate.

the comparator with DDNR, 25% in the DAC, 11% in the logic, and 2% in the S&H.

To validate the DDNR method, V_{bias} (which controls the reference delay) was swept to obtain the results in Fig. 19, similar to the theoretical curves in Fig. 5. With increasing V_{bias} , the input-referred noise gradually decreases from 0.95LSB down to 0.66LSB. This is in close agreement to the theoretical improvement for $5\times$ voting, as shown in Fig. 19. While the reduction from 0.95LSB to 0.66LSB does not seem too impressive, one should realize that $4\times$ analog scaling of the comparator would be needed to achieve an equivalent improvement, as explained in Section II. Fig. 19 also shows the measured power consumption, which starts at 74 nW and gradually increases with increasing V_{bias} . For the other measurements in this paper, V_{bias} was set to 0.415 V as indicated by a circle in the figure. At that point, the power consumption equals 96 nW, which is 22 nW higher as compared to the case where majority voting is inactive. The post-layout simulated comparator consumes 90 fJ per comparison. Thus, as the ADC operates at 40 kS/s, this 22 nW leads to an estimate that 6 additional comparisons per conversion are performed for the majority-voting scheme. This is in line with the expectation in Fig. 5, where also about 6 additional comparisons are needed for the same scenario ($\alpha = 1$).

In the end, the achieved ENOB (10.1 bit, Fig. 17) is as expected, which can be seen as follows: in the selected case ($V_{bias} = 0.415$ V), the noise is reduced to 0.71 LSB (Fig. 19), which corresponds to an SNR of 66 dB. When using an ADC model in MATLAB that includes the non-linearity from the measured INL (Fig. 15), a THD close to -66 dB is predicted. When combining the THD and SNR, an SNDR of 63 dB can be estimated, leading to an expected ENOB of 10.2 bit. This is indeed close to the measured result of 10.1 bit.

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Fig. 19. Measured ADC input-referred noise and measured power consumption versus V_{bias} in 12 bit mode and at 40 kS/s.TABLE I
PERFORMANCE AT DIFFERENT SUPPLY VOLTAGES

Mode	10bit				12bit			
Supply (V)	0.50	0.55	0.60	0.70	0.50	0.55	0.60	0.70
Sample rate (S/s)	10k	20k	40k	100k	10k	20k	40k	100k
ENOB @ Nyq. (bit)	8.69	9.07	9.40	9.53	9.12	9.54	10.09	10.28
Power (nW)	12.1	29.6	71.7	246	13.5	32.7	96.8	331
FOM (fJ/conv.step)	2.9	2.8	2.7	3.3	2.4	2.2	2.2	2.7

TABLE II
PERFORMANCE SUMMARY AND COMPARISON

	[12]	[1]	[8]	[2]	This work	
Technology	65nm	90nm	180nm	90nm	65nm	
Area (mm ²)	0.026	0.03	0.086	0.047	0.076	
Supply voltage (V)	1	0.35	1	0.7	0.6	
Sample rate (S/s)	1M	100k	10M	2M	40k	
Resolution (bit)	10	10	10	7 ~ 10	10	12
INL (LSB)	2.2	0.6	0.38	0.16 ~ 0.68	0.48	1.90
DNL (LSB)	0.5	0.3	0.34	0.31 ~ 0.50	0.32	0.97
ENOB @ Nyquist (bit)	8.75	9.05	9.83	6.94 ~ 9.30	9.4	10.1
Power (μ W)	1.9	0.17	98	1.61 ~ 3.56	0.072	0.097
FOM (fJ/conv.step)	4.4	3.2	11	6.6 ~ 2.8	2.7	2.2

Table I shows the measured results of this chip for various supply voltages below and above the nominal supply of 0.6 V. The maximum sample rate scales with the supply from 10 kS/s at 0.50 V up to 100 kS/s at 0.70 V. The ENOB degrades at lower supplies due to a reduction of the SNR, but mostly due to increased distortion from the S&H. Nonetheless, as the power consumption also scales down rapidly, the FOM can be maintained between 2.2 and 3.3 fJ/conversion-step in all settings. Note that V_{bias} was only optimized for the nominal supply of 0.6 V, thus some improvement is still feasible for the other supply voltages.

Table II summarizes the measured performance and gives a comparison to prior art. Compared to low-power SAR ADCs,

the power efficiency of 2.2 fJ/conversion-step as well as the ENOB of 10.1 bit are better than previous work.

V. CONCLUSION

In this work, an ultra low-power 10/12 bit 40 kS/s SAR ADC for sensor applications was presented. The ADC applies a self-oscillating comparator, a segmented DAC with 250 aF unit elements and a Data-Driven Noise-Reduction method. The DDNR method enables better power efficiencies at higher resolutions, as the comparator noise is selectively reduced. The power-efficiency of the implemented prototype in 65 nm CMOS is 2.2 fJ/conversion-step with an ENOB of 10.1 bit.

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Pieter Harpe (M'12) received the M.Sc. and Ph.D. degrees from the Eindhoven University of Technology, The Netherlands.

In 2008, he started as researcher at Holst Centre/imec, The Netherlands. Since then, he has been working on ultra low-power wireless transceivers, with a main focus on ADC research and design. In April 2011, he joined Eindhoven University of Technology as an Assistant Professor of low-power mixed-signal circuits.

Since 2012, Dr. Harpe has been co-organizer of the yearly workshop on Advances in Analog Circuit Design (AACD). In 2013, he became a member of the Technical Program Committees of ISSCC and ESSCIRC. In 2013, he received a VENI grant from the Dutch Technology Foundation STW for research on multi-functional ADCs.



Eugenio Cantatore (M'98) received the Master and Ph.D. degrees in electrical engineering from Politecnico di Bari, Bari, Italy, in 1993 and 1997, respectively. From 1994 to 1999, he was first a Ph.D. student and then a fellow at the European Laboratory for Particle Physics (CERN) Geneva, Switzerland.

In 1999 he moved to Philips Research, Eindhoven, The Netherlands, as a Senior Scientist and in 2007 joined the Eindhoven University of Technology where he is an Associate Professor. His research interests include the design and characterization of electronic circuits exploiting emerging technologies as well as the design of ultra low-power micro-systems for medical applications. He has authored or co-authored more than 110 papers in journals and conference proceedings, and has 13 patents or patent applications.

Dr. Cantatore is active in the Technical Program Committees of ESSDERC, ESSCIRC and ISSCC and has been a member of the Executive Committee of ISSCC. Since 2013 he has been Chair of the Technology Directions subcommittee of ISSCC. In 2006 he received from ISSCC the Beatrice Winner Award for Editorial Excellence and was nominated in the Scientific American top 50 list. He received the Philips Research Invention Award in 2007 and in 2012 the Best Paper Award from ESSDERC.



Arthur H. M. van Roermund (SM'95) was born in Delft, The Netherlands, in 1951. He received the M.Sc. degree in electrical engineering in 1975 from Delft University of Technology and the Ph.D. degree in applied sciences from K. U. Leuven, Belgium, in 1987.

From 1975 to 1992, he was with Philips Research Laboratories in Eindhoven. From 1992 to 1999 he has been a full Professor in the Electrical Engineering Department of Delft University of Technology, where he was Chairman of the Electronics Research Group and a member of the management team of DIMES. From 1992 to 1999, he was Chairman of a two-years post-graduate school for "chartered designer". From 1992 to 1997, he was a consultant for Philips. In October 1999, he joined Eindhoven University of Technology as a full Professor, chairing the Mixed-signal Microelectronics Group. From September 2002 to September 2012, he was Director of Research of the Department of Electrical Engineering. From 2009 to 2012, he was a member of the supervisory board of the NRC Photonics Research Centre. He is Chairman of the Board of ProRISC, the microelectronics platform in The Netherlands, and Vice Chair of the ICT-research platform for The Netherlands (IPN). He has authored/co-authored about 500 articles and 30 books.

Since 2001, Dr. van Roermund has been standing co-organizer of the yearly workshop on Advanced Analog Circuit Design (AACD). In 2004, he received the Simon Stevin Meester Award for his scientific and technological achievements. In 2007 he was a member of an international assessment panel for the Department of Electronics and Information of Politecnico di Milano, Italy; in 2009 for Electronics and Electrical Engineering for the merged Aalto University Finland; and in 2012 for KTH, Stockholm, Sweden.